DDCS V4.1 · SEA OTTER

User manual

Replacement controller software for the DDCS V4.1 — daily use at the machine



github.com/JSFrouws/ddcs-4.1-Sea-Otter · docs/user-manual.md

For the person at the machine. Sea Otter is the replacement controller software for the DDCS V4.1 (`app/`). This manual covers daily use. Design background is in operator-procedures.md, the network API in api-reference.md.

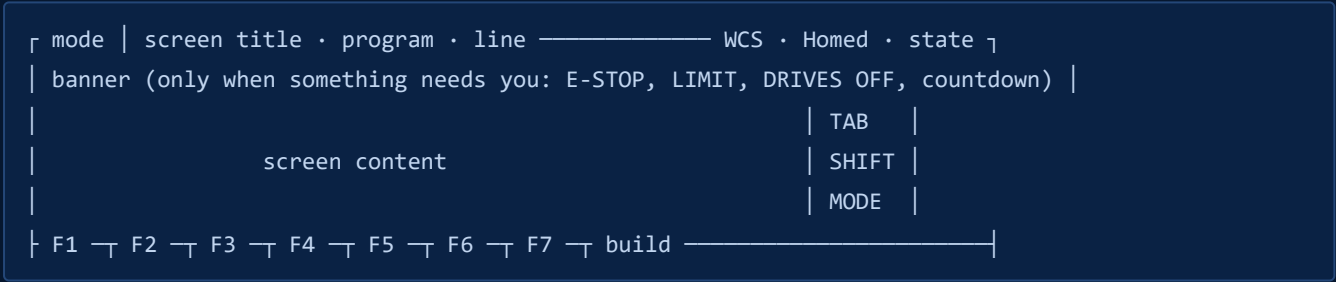
> **Not yet tested on a machine with drives connected (2026-10-21).** Tested on a real controller with no drives or motors attached: step-exact positions, real CAM programs up to 64 000 lines, limit inputs and controlled stops through the real input path. Do the first real tests with the motors unpowered or able to move freely, with a hard-wired E-stop that cuts the drives. No warranty, no liability: see the README.

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0. Installing

From a USB stick, no PC tools needed on the controller: usb-install.md. In short: copy the `ddcsv4` install folder to a FAT32 stick, set the IP address in `network.conf`, plug it in, switch on, then take the stick out. The browser panel and the API answer at `http://<that IP>:8080/`. To go back to the vendor software, see the same page. Remote access with a password: ssh-setup.md.

1. The screen



- **Top right**: active work offset (G54…), `Homed ✓/–`, and the machine **state** (READY, RUN, HOLD, E-STOP, LIMIT, DRIVES OFF, SIMULATE…).
- **Soft keys F1–F7** along the bottom: their labels change per screen. **F7 is always Page R** (next screen); **Shift then F7** goes back.
- **Right strip**: what **Tab**, **Shift** and **Mode** do right now.
- **Toasts** (white box near the bottom): the result of what you just did, or why a key was refused. Nothing is refused silently.

2. Keys

Key	Does
Mode	Run ↔ Jog. Jog is locked while a program runs or holds.
F7 Page R	Run → Files → Work → Tools → Macros → Settings → Run
Tab	Next pane on the screen (e.g. Toolpath ↔ Program)
Shift	ALT layer for the next key (lit while active): touch-off, Page R back, …
Y± / X±	Move in lists and tables; jog in Jog
Enter / Select / Modify	Open / accept / edit the focused item
Cancel	Back / discard
F1 Start	Start or resume the program
F2 Pause	Feed hold on **every** screen while something moves
F3 Reset	Stop on **every** screen. Also switches the drives on after DRIVES OFF

3. Daily procedure

3.1 Power up

1. Switch on. If the E-stop is pressed, release it. The banner tells you what's needed.
2. **Mode** → Jog. **F1 Home** if you want soft limits (not required). Reset stops homing.

3.2 Load a program

1. **Page R** → Files. **F3** switches Local / USB. **Y±** picks a file.
2. **F1 Load** (or Select). The whole file is checked first; an unsupported code is reported with its line number and the file is not loaded. Supported: G0–G3 (also in the G17/G18/G19 planes), G4, G10 L2/L20, G20/G21, G28, G53, G54–G59, drilling cycles G73/G81/G82/G83/G85/G89 with G80 and G98/G99, G90/G91, G94, M0/M1, M3/M4/M5, M6 (pause for a tool change), M7/M8/M9, M10/M11, M30. G40/G49/G61/G64 are accepted. G43 is accepted but ignored (no tool length compensation). G41/G42 (cutter compensation) and G84 (tapping) are refused.

3.3 Zero the workpiece

1. Jog to the part's edge (Jog: arrow keys; **F4 Step** picks 0.01 / 0.1 / 1 / 10 mm or continuous).
2. **Shift + the axis key towards the part** sets that axis: with an edge Ø set (Jog setup, or Tools › F3) it accounts for the radius (X+ edge → X = −Ø/2). Press the same key again to set 0 instead.
3. Z: touch the top, **Shift + Z−** → Z = 0.
4. Check with **Goto 0** (Jog F2): Z up to safe height, then XY to 0.
5. Several parts: **Jog F5 WCS** picks G55, G56…; on **Work** mark them **F4 In job** and Start runs the program on each one in turn.

3.4 Check, then run

F5 TryCut on Run cycles what Start will do:

Armed	What happens
Run	The real job
TryCut	Spindle off; Z goes once to the **clearance level** (work Z0 + Safe Z) and stays there. The machine flies the whole path in the air
Simulate	Nothing moves. The position readout turns **orange**: those coordinates are fictitious. Afterwards the real position is shown again. M0/M1/M6 pauses still stop it, just like a real run

F1 Start. A program started over the network counts down on the panel first (default 5 s); **Reset cancels**.

3.5 During the run

- **Overrides**: pendant switch OFF + hold the dead-man + turn the wheel: **X1 = feed**, **X10 = spindle** (2 % per click), **X100 = feed in 10 % steps**. Feed and spindle go up to **300 %**. The bars stop at 200 % and the number shows the rest in amber. **Rapid follows the feed down** and never goes above 100 %. Its bar is half width, so 100 % lines up on all three bars.
- **The speed is capped** at what the controller can actually deliver (960 mm/min per axis at 2000 steps/mm). A higher F or override runs at the cap.
- **F4 View** cycles **2D → ISO → CPU saver**. CPU saver hides the toolpath and redraws once a second, leaving more time for motion. After an E-stop or limit the view switches to ISO by itself.
- **M6 / M0**: the machine holds with a message ("change to T3, then Start"). Do the change, **F1 Start** continues.
- **F2 Pause** = feed hold (F1 resumes). **F3 Reset** stops; the lines done so far are remembered.

3.6 Resume after a stop

Run → **Tab** → Program. Pick the line (**Y±** one line, **X±** ten), **Select**. The bar shows what Start will restore (WCS, F, S, tool, coolant) and how it approaches: Z up, XY across, then down. **F1 Start**.

4. When the machine stops on its own

4.1 Controlled stop (no lost steps)

Settings › Homing & limits › Controlled stop on Reset, E-stop, limit — **on by default**. Reset, the E-stop and a limit switch all **brake along the path** at the axes' maximum deceleration, so the motors don't lose steps and the position stays valid: no re-homing, no re-zeroing. The machine runs on by the **braking distance**: about **1 mm** at full speed (measured). **Mount limit switches with at least that much room before the hard stop.** Switch it off and the stop is abrupt (≈ 0.3–0.5 mm), but a fast-moving motor may then lose steps.

4.2 E-stop

The E-stop is for **unexpected or unwanted movement**. It stops motion, spindle and outputs. **The drives stay on**, so the axes keep their position and holding torque. Release the button and the machine is ready again. Jog, or resume from a line. **To remove power, switch the machine off.**

4.3 Limit switch

The banner shows e.g. `X+ LIMIT`. Only that direction is blocked. **Mode** → Jog and:

- **back off** the switch (X− here): the limit clears by itself when the switch releases;
- or move **sideways** (Y) or **up/down** (Z): never blocked.

Two or three switches at once (e.g. driving into a corner): each one stops the move and blocks its own direction; the banner lists them all (`X+ Y- LIMIT · jog X- Y+ to release`). LIMIT only goes away when the **last** switch is released.

A program that was running has stopped with its progress remembered: resume as in 3.6. Which input each axis's switch is on: *Settings › Homing & limits › X/Y/Z limit/home switch input* (this machine: IN02 / IN03 / IN04; 0 = no switch on that axis).

4.3a Soft limits and bench limits

Two software boxes keep the machine inside its travel, without touching a switch:

	In force	Set in *Settings › Homing & limits*
Soft limits	once the machine is **homed** (before that, positions mean nothing)	*X/Y/Z soft limit − / +*, switch: *Soft limits*
Bench limits	**always**, homed or not: a work box, e.g. the part of the table you use, or a safe box on the bench while testing	*X/Y/Z bench limit − / +*, switch: *Bench limits*

When both are on, the tighter one counts. What they do:

- **A program that would leave the box is refused** at Start, with the axis, the side and how far: e.g. *Program goes past the X+ bench limit on G54: reaches 675.8 mm, limit 587.8*. TryCut is checked at its clearance height. Simulate still runs (it moves nothing) and says the program is outside.
- **A jog towards the edge slows down and stops exactly on it.** A step jog past the edge is shortened to end on the edge; a held (continuous) jog brakes at the axis' acceleration as it gets close. Measured on the controller: from 16 mm/s to a standstill within the last ~0.7 mm, landing on the edge to the step. The next jog that way is refused (*X+ is at its bench limit*); the other way is free.
- Goto 0, Park and Tool change are refused if their target is outside the box.

4.4 DRIVES OFF

Only when someone switched the drives off (Jog F3, or over the network). The banner says who. **F3 Reset** on any screen switches them on again.

5. Macros

Page R → Macros. **Y±** pick, **Select / Enter / F1** run. **Reset** stops any macro.

Macro	Does
Safe Z	Z up to work Z0 + Safe Z (only ever up)
Go to work zero	Z up, then X Y to the active work zero
Park	Z up, X Y to the park position, Z to its height *(homed machine only)*
Tool change position	Same, to the tool-change position *(homed machine only)*
Spindle warm-up	Spindle at the warm-up speed for the warm-up time, then off
Zero all	X Y Z of the active work offset = the current position

Positions, speed and time: *Settings › Macros* (machine coordinates).

6. Settings

Page R → Settings. Left: groups, right: values. Quick press / Modify / Enter starts editing. On/off, lists and small numbers change in place with Y± (or hold + wheel); larger numbers open the digit editor. **Enter accepts, Cancel discards.** Moving past the first or last value goes back to the groups, so the pendant alone can do everything.

- **Grey values don't apply** right now, and the reason is shown (e.g. "API is off"). They can't be edited until the setting they depend on is changed.
- Values marked **idle only** (machine settings) can't change while anything moves.
- **F2** restores this machine's original values. **F4** exports the settings to USB.

Useful ones: *Display › Default toolpath view* (2D / ISO / CPU saver) · *Jog & touch-off › Safe Z* · *Homing & limits › Controlled stop* · *Network & API › Allow remote START* (on by default; switch off if you don't want a network client to start programs) · *Pendant › Reverse wheel direction*.

7. The pendant (optional)

Input	Does
Wheel, dead-man released	**Moves the focus**: list items, settings, digit cursor, view
Wheel, dead-man held, switch OFF	**Changes** the focused value; while running: overrides
Wheel, dead-man held, switch X/Y/Z/4	**Jogs** that axis 0.001 / 0.01 / 0.1 mm per count (refused while running)
Quick press of the dead-man (switch OFF)	**Enter** on the focused item
Pendant E-stop	Same as the panel E-stop (§ 4.2)

8. Network

The controller serves a browser panel and the API on port 8080 (`http://<controller-ip>:8080/`). Every panel action is also an API call; see api-reference.md. Programs can be uploaded and started remotely. A remote start always counts down on the panel first, and Reset cancels it.

9. Known limits (2026-10-21)

- **Not yet tested with drives or motors connected**, nor with real limit/home switches or a probe (probing is not configured). Limit handling was tested through the real input path with the inputs' level inverted, not with a switch.
- Loading a very large file (64 000 lines) takes ~4 s, and the panel freezes for that time.
- G84 tapping and tool length compensation (G43) are not supported.
- `G28` without axis words sends all axes to machine zero **together**, not Z first, as written in the program. Check CAM output before the first real run.